

Supporting Information

When do kinase inhibitors change protonation or tautomeric state upon binding?

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Detail of Methods Used

Electrostatic free energy in MCCE

DelPhi [1-4] provides numerical solutions to the Poisson Boltzmann equation to determine the electrostatic interactions. A low dielectric region is placed around the protein with the program IPECE [5] oriented to cover a 30 Å thick hydrophobic region of the protein that would be embedded in the membrane hydrocarbon region. The dielectric constant of four is used for all atoms including the membrane. A dielectric constant of 80 is used for the external solvent and for internal cavities with a radius >1.4 Å. The salt concentration is 150 mM with a Stern layer of 2 Å. Thus, the dielectric response in MCCE includes both the implicit dielectric constant used in DelPhi and explicit response allowed by side chain and ligand conformers which sample of multiple dipole orientation. The Amber force field [6], used to calculate the van der Waals interactions, is reduced to one eighth of the standard value [7].

Microstate energy used in Monte Carlo sampling

The pre-made conformers of all residues, cofactors (inhibitors) are subjected to Monte Carlo (MC) sampling to generate a Boltzmann distribution of conformers. A microstate consists of one conformer per residue or cofactor. Metropolis-Hastings Sampling is performed to determine its acceptance [8]. The energy ΔH^x of microstate x is:

$$\begin{aligned} \Delta H^x = & \sum_{i=1}^M \delta_{x,i} \left\{ \left[2.3 m_i k_B T (\text{pH} - \text{p}K_{a,\text{sol},i}) + n_i F (E_h - E_{m,\text{sol},i}) \right] \right. \\ & + \left[\Delta \Delta G_{\text{solv},i} + \Delta G_{\text{bkbn},i}^{\text{CE}} + \Delta G_{\text{bkbn},i}^{\text{LJ}} + \Delta G_{\text{torsion},i} + \Delta \Delta G_{\text{SAS},i} \right] \\ & \left. + \sum_{j=i+1}^M \delta_{x,i} \left[\Delta G_{ij}^{\text{CE}} + \Delta G_{ij}^{\text{LJ}} \right] \right\} \end{aligned} \quad (1)$$

where, M is the total number of conformers. The first line in eq S1 describes the electrochemical properties of each conformer, where $\delta_{x,i}$ is 1 if conformer i is present in the microstate or 0 otherwise. The ability of the group to participate in acid/base reactions is described for protons in the first term and redox reactions in the second term. m_i is 1 for protonated bases, -1 for deprotonated acids, and 0 for all neutral conformers. $pKa_{sol,i}$ is the reference pKa of this residue type in solution, and $Em_{sol,i}$ is the reference electrochemical midpoint potential of this redox active group.

k_B is the Boltzmann constant, T is temperature in Kelvin (default is 300 K), F is the Faraday constant, and n_i is the change in the number of electrons on a conformer type that is taking part in a redox titration. pH and Eh are the relevant solution parameters describing the chemical potential of protons or electrons that will come to equilibrium with the protein. The second line of eq S1 describes the conformer self-energies of the microstate, which are independent of the other conformer choices of other residues and cofactors. These energies are the loss of the conformer’s solvation energy as it moves from the referenced solvent to its position in the protein ($\Delta G_{sol,i}$), continuum electrostatic (CE) and Lennard-Jones (LJ) van der Waals interactions with the fixed backbone amides, torsion energy and the favorable LJ interactions with the exposed surface of the cofactor/residual sidechain with the implicit solvent, respectively [9]. $\Delta G_{sol,i}$ is set to zero here. The third line gives the CE and LJ pairwise interactions between the conformer and all other conformers selected in the microstate.

Inhibitors

Figure S1: Inhibitor Conformer States and Ensemble Charges

(2) = two protons on that atom. Ensemble charges from MCCE4 at pH 7.0.

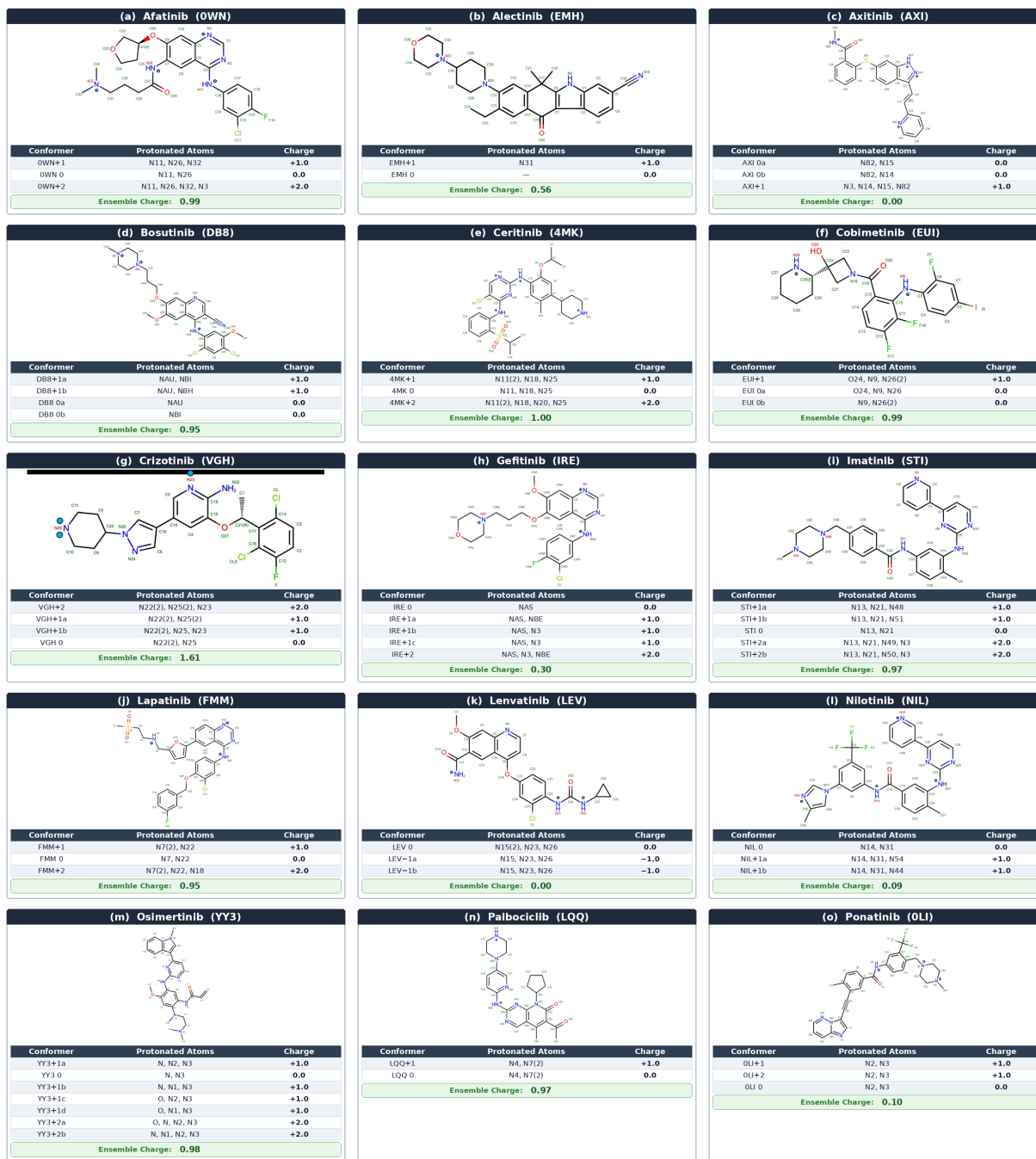


Figure S1: Inhibitor conformer definitions and ensemble charges. 2D chemical structures and protonation/tautomer states sampled for each kinase inhibitor in MCCE calculations. Atom labels follow PDB naming conventions; labile nitrogen atoms are labeled in red. Suffix notation (e.g., +1a, +1b) distinguishes tautomers of the same net charge; (2) denotes two protons on that atom. Ensemble charges are Boltzmann-weighted averages from Monte Carlo sampling of the isolated inhibitor in solution at pH 7.4. Part 1: (a) Afatinib, (b) Alectinib, (c) Axitinib, (d) Bosutinib, (e) Ceritinib, (f) Cobimetinib, (g) Crizotinib, (h) Gefitinib, (i) Imatinib.

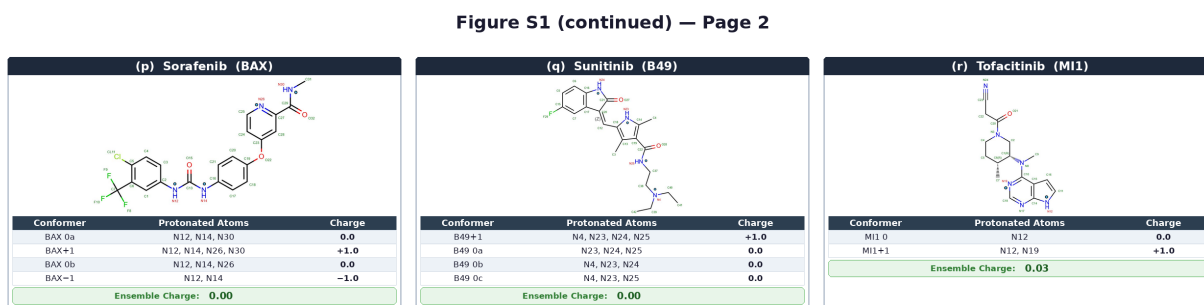


Figure S1 (continued): Inhibitor conformer definitions and ensemble charges.

Table S1: Residue ranges used in kinase calculations. 1st res and lst res: first and last residue numbers in the PDB file. Cap 1st and cap lst: first and last residue numbers used for capping where applied. Empty cap entries indicate no capping was used.

Kinase	PDB ID	1st res	last res	cap 1st	cap last
<i>ABL: ABL1 tyrosine protein kinase</i>					
ABL	3UE4	233	502	245	497
ABL	2HYY	235	498	245	497
ABL	3PYY	251	519	245	497
ABL	3CS9	233	500	245	497
ABL	5MO4	83	531	245	497
<i>ALK: Anaplastic lymphoma kinase</i>					
ALK	3AOX	1086	1401	—	—
ALK	4MKC	1093	1401	—	—
ALK	2XP2	1093	1401	—	—
ALK	2YFX	1093	1401	—	—
ALK	4ANQ	—	—	—	—
ALK	4ANS	—	—	—	—
ALK	5AAA	—	—	—	—
ALK	5AAB	—	—	—	—
<i>CDK6: Cyclin-dependent kinase 6</i>					
CDK6	2EUF	—	—	—	—
CDK6	5L2I	10	301	—	—
<i>DDR1: Discoidin domain receptor tyrosine kinase</i>					
DDR1	3ZOS (buried)	602	913	—	—
DDR1	3ZOS (surface)	602	913	—	—
<i>EGFR: Epidermal growth factor receptor kinase</i>					

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Kinase	PDB ID	1st res	last res	cap 1st	cap last
EGFR	4G5J	696	1016	703	982
EGFR	4G5P	696	1016	703	982
EGFR	2ITO	696	1019	703	982
EGFR	2ITY	697	1019	703	982
EGFR	2ITZ	—	—	—	—
EGFR	3UG2	—	—	—	—
EGFR	4I22	—	—	—	—
EGFR	4WKQ	—	—	—	—
EGFR	1XKK	702	1018	703	982
EGFR	4ZAU	698	1017	703	982
<i>JAK3: Janus kinase 3</i>					
JAK3	3LXK	815	1098	—	—
<i>MEK: Mitogen-activated protein kinase kinase</i>					
MEK	4AN2	58	381	—	—
MEK	4LMN	61	382	—	—
<i>MET: Tyrosine protein kinase Met</i>					
MET	2WGJ	1052	1345	1067	1345
<i>VEGFR: Vascular endothelial growth factor receptor kinase</i>					
VEGFR	4AG8	816	1168	—	—
VEGFR	4AGC	801	1169	—	—
VEGFR	3WZD	820	1167	—	—
VEGFR	3WZE	815	1167	—	—
VEGFR	4ASD	807	1168	—	—
VEGFR	4AGD	802	1169	—	—

Table S2: Kinase inhibitor conformer charge states and MCCE calculated occupancies. Results grouped by kinase type in bold. For each inhibitor (PDB structure) in italics, inhibitor conformer protonated atoms, Epik Energy (kcal/mol), Charge, $P(i)_{\text{soln}}$: MCCE calculated distribution in solution, $P(i)_{\text{prot}}$: MCCE calculated distribution in bound form, and the MCCE conformer name are listed. The bold **Ensemble** row reports the MCCE Monte Carlo Boltzmann-weighted mean charge for both solution and protein environments.

Conformer	Protonated atoms	Energy	Charge	$P(i)_{\text{soln}}$	$P(i)_{\text{prot}}$	MCCE
ABL: ABL1 tyrosine protein kinase						
<i>Bosutinib (3UE4)</i>						
DB8 0a	NAU	1.715	0.0	0.055	0.046	DB801
DB8 0b	NBI	5.065	0.0	0.000	0.000	DB802
DB8+1a	NAU, NBI	0.071	1.0	0.887	0.560	DB8+1
DB8+1b	NAU, NBH	1.695	1.0	0.057	0.394	DB8+2
Ensemble			0.944	0.944	0.954	

<i>Imatinib (2HYY)</i>						
STI 0	N13, N21	2.031	0.0	0.032	0.000	STI01
STI+1a	N13, N21, N48	0.316	1.0	0.587	0.002	STI+1
STI+1b	N13, N21, N51	0.573	1.0	0.380	0.998	STI+2
STI+2a	N13, N21, N49, N3	5.430	2.0	0.000	0.000	STI+a
STI+2b	N13, N21, N51, N3	5.687	2.0	0.000	0.000	STI+b
Ensemble			0.968	0.967	1.000	

<i>Imatinib (3PYY)</i>						
STI 0	N13, N21	2.031	0.0	0.032	0.000	STI01
STI+1a	N13, N21, N48	0.316	1.0	0.587	0.004	STI+1
STI+1b	N13, N21, N51	0.573	1.0	0.380	0.996	STI+2
STI+2a	N13, N21, N49, N3	5.430	2.0	0.000	0.000	STI+a
STI+2b	N13, N21, N51, N3	5.687	2.0	0.000	0.000	STI+b
Ensemble			0.968	0.967	1.000	

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Conformer	Protonated atoms	Energy	Charge	$P(i)_{\text{soln}}$	$P(i)_{\text{prot}}$	MCCE
<i>Nilotinib (3CS9)</i>						
NIL 0	N14, N31	0.058	0.0	0.907	0.907	NIL01
NIL+1a	N14, N31, N54	1.404	1.0	0.093	0.093	NIL+1
NIL+1b	N14, N31, N44	5.173	1.0	0.000	0.000	NIL+2
Ensemble			0.094	0.093	0.093	

<i>Nilotinib (5MO4)</i>						
NIL 0	N14, N31	0.058	0.0	0.907	0.777	NIL01
NIL+1a	N14, N31, N54	1.404	1.0	0.093	0.223	NIL+1
NIL+1b	N14, N31, N44	5.173	1.0	0.000	0.000	NIL+2
Ensemble			0.094	0.093	0.223	
ALK: Anaplastic lymphoma kinase						
<i>Alectinib (3AOX)</i>						
EMH 0	—	0.488	0.0	0.439	0.257	EMH01
EMH+1	N31	0.342	1.0	0.561	0.743	EMH+1
Ensemble			0.561	0.561	0.743	

<i>Ceritinib (4MKC)</i>						
4MK 0	N11, N18, N25	3.967	0.0	0.000	0.000	4MK01
4MK+1	N11(2), N18, N25	0.000	1.0	1.000	1.000	4MK+1
4MK+2	N11(2), N18, N20, N25	5.439	2.0	0.000	0.000	4MK+a
Ensemble			0.999	1.000	1.000	

<i>Crizotinib (2XP2)</i>						
VGH 0	N22(2), N25	3.192	0.0	0.005	0.000	VGH01
VGH+1a	N22(2), N25(2)	0.585	1.0	0.373	1.000	VGH+1
VGH+1b	N22(2), N25, N23	3.587	1.0	0.002	0.000	VGH+2
VGH+2	N22(2), N25(2), N23	0.283	2.0	0.620	0.000	VGH+a

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Conformer	Protonated atoms	Energy	Charge	$P(i)_{\text{soln}}$	$P(i)_{\text{prot}}$	MCCE
Ensemble			1.616	1.615	1.000	
<i>Crizotinib (2YFX)</i>						
VGH 0	N22(2), N25	3.192	0.0	0.005	0.000	VGH01
VGH+1a	N22(2), N25(2)	0.585	1.0	0.373	1.000	VGH+1
VGH+1b	N22(2), N25, N23	3.587	1.0	0.002	0.000	VGH+2
VGH+2	N22(2), N25(2), N23	0.283	2.0	0.620	0.000	VGH+a
Ensemble			1.616	1.615	1.000	
<i>Crizotinib (4ANQ)</i>						
VGH 0	N22(2), N25	3.192	0.0	0.005	0.000	VGH01
VGH+1a	N22(2), N25(2)	0.585	1.0	0.373	1.000	VGH+1
VGH+1b	N22(2), N25, N23	3.587	1.0	0.002	0.000	VGH+2
VGH+2	N22(2), N25(2), N23	0.283	2.0	0.620	0.000	VGH+a
Ensemble			1.616	1.615	1.000	
<i>Crizotinib (4ANS)</i>						
VGH 0	N22(2), N25	3.192	0.0	0.005	0.000	VGH01
VGH+1a	N22(2), N25(2)	0.585	1.0	0.373	1.000	VGH+1
VGH+1b	N22(2), N25, N23	3.587	1.0	0.002	0.000	VGH+2
VGH+2	N22(2), N25(2), N23	0.283	2.0	0.620	0.000	VGH+a
Ensemble			1.616	1.615	1.000	
<i>Crizotinib (5AAA)</i>						
VGH 0	N22(2), N25	3.192	0.0	0.005	0.000	VGH01
VGH+1a	N22(2), N25(2)	0.585	1.0	0.373	1.000	VGH+1
VGH+1b	N22(2), N25, N23	3.587	1.0	0.002	0.000	VGH+2
VGH+2	N22(2), N25(2), N23	0.283	2.0	0.620	0.000	VGH+a
Ensemble			1.616	1.615	1.000	

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Conformer	Protonated atoms	Energy	Charge	$P(i)_{\text{soln}}$	$P(i)_{\text{prot}}$	MCCE
<i>Crizotinib (5AAB)</i>						
VGH 0	N22(2), N25	3.192	0.0	0.005	0.000	VGH01
VGH+1a	N22(2), N25(2)	0.585	1.0	0.373	1.000	VGH+1
VGH+1b	N22(2), N25, N23	3.587	1.0	0.002	0.000	VGH+2
VGH+2	N22(2), N25(2), N23	0.283	2.0	0.620	0.000	VGH+a
Ensemble			1.616	1.615	1.000	
CDK6: Cyclin dependent kinase 6						
<i>Palbociclib (2EUF)</i>						
LQQ 0	N4, N7(2)	2.434	0.0	0.025	0.000	LQQ01
LQQ+1	N4, N7	0.256	1.0	0.975	1.000	LQQ+1
Ensemble			0.975	0.975	1.000	
<i>Palbociclib (5L2I)</i>						
LQQ 0	N4, N7(2)	2.434	0.0	0.025	0.006	LQQ01
LQQ+1	N4, N7	0.256	1.0	0.975	0.994	LQQ+1
Ensemble			0.975	0.975	0.994	
DDR1: Discoidin domain receptor tyrosine kinase						
<i>Ponatinib (3ZOS, buried[†])</i>						
0LI 0	N2	2.030	0.0	0.033	0.000	0LI01
0LI+1	N2, N3	0.316	1.0	0.587	0.000	0LI+1
0LI+2	N2, N4	0.573	1.0	0.381	1.000	0LI+2
Ensemble			0.967	0.968	1.000	
<i>Ponatinib (3ZOS, surface[†])</i>						
0LI 0	N2	2.030	0.0	0.033	0.118	0LI01
0LI+1	N2, N3	0.316	1.0	0.587	0.193	0LI+1
0LI+2	N2, N4	0.573	1.0	0.381	0.689	0LI+2
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Conformer	Protonated atoms	Energy	Charge	$P(i)_{\text{soln}}$	$P(i)_{\text{prot}}$	MCCE
Ensemble			0.967	0.968	0.882	
EGFR: Epidermal growth factor receptor kinase						
<i>Afatinib (4G5J)</i>						
OWN 0	N11, N26	2.861	0.0	0.007	0.050	OWN01
OWN+1	N11, N26, N32	0.005	1.0	0.993	0.950	OWN+1
OWN+2	N11, N26, N32, N3	4.147	2.0	0.000	0.000	OWN+a
Ensemble			0.993	0.993	0.950	
<i>Afatinib (4G5P)</i>						
OWN 0	N11, N26	2.861	0.0	0.007	0.038	OWN01
OWN+1	N11, N26, N32	0.005	1.0	0.993	0.962	OWN+1
OWN+2	N11, N26, N32, N3	4.147	2.0	0.000	0.000	OWN+a
Ensemble			0.993	0.993	0.962	
<i>Gefitinib (2ITO)</i>						
IRE 0	NAS	0.212	0.0	0.700	0.300	IRE01
IRE+1a	NAS, NBE	0.717	1.0	0.298	0.700	IRE+1
IRE+1b	NAS, N3	3.728	1.0	0.002	0.000	IRE+2
IRE+1c	NAS, N3	4.395	1.0	0.000	0.000	IRE+3
IRE+2	NAS, N3, NBE	4.412	2.0	0.000	0.000	IRE+a
Ensemble			0.302	0.300	0.700	
<i>Gefitinib (2ITY)</i>						
IRE 0	NAS	0.212	0.0	0.700	0.634	IRE01
IRE+1a	NAS, NBE	0.717	1.0	0.298	0.366	IRE+1
IRE+1b	NAS, N3	3.728	1.0	0.002	0.000	IRE+2
IRE+1c	NAS, N3	4.395	1.0	0.000	0.000	IRE+3
IRE+2	NAS, N3, NBE	4.412	2.0	0.000	0.000	IRE+a
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Conformer	Protonated atoms	Energy	Charge	$P(i)_{\text{soln}}$	$P(i)_{\text{prot}}$	MCCE
Ensemble			0.302	0.300	0.366	
<i>Gefitinib (2ITZ)</i>						
IRE 0	NAS	0.212	0.0	0.700	0.320	IRE01
IRE+1a	NAS, NBE	0.717	1.0	0.298	0.680	IRE+1
IRE+1b	NAS, N3	3.728	1.0	0.002	0.000	IRE+2
IRE+1c	NAS, N3	4.395	1.0	0.000	0.000	IRE+3
IRE+2	NAS, N3, NBE	4.412	2.0	0.000	0.000	IRE+a
Ensemble			0.302	0.300	0.680	
<i>Gefitinib (3UG2)</i>						
IRE 0	NAS	0.212	0.0	0.700	0.986	IRE01
IRE+1a	NAS, NBE	0.717	1.0	0.298	0.014	IRE+1
IRE+1b	NAS, N3	3.728	1.0	0.002	0.000	IRE+2
IRE+1c	NAS, N3	4.395	1.0	0.000	0.000	IRE+3
IRE+2	NAS, N3, NBE	4.412	2.0	0.000	0.000	IRE+a
Ensemble			0.302	0.300	0.014	
<i>Gefitinib (4I22)</i>						
IRE 0	NAS	0.212	0.0	0.700	0.590	IRE01
IRE+1a	NAS, NBE	0.717	1.0	0.298	0.410	IRE+1
IRE+1b	NAS, N3	3.728	1.0	0.002	0.000	IRE+2
IRE+1c	NAS, N3	4.395	1.0	0.000	0.000	IRE+3
IRE+2	NAS, N3, NBE	4.412	2.0	0.000	0.000	IRE+a
Ensemble			0.302	0.300	0.410	
<i>Gefitinib (4WKQ)</i>						
IRE 0	NAS	0.212	0.0	0.700	0.506	IRE01
IRE+1a	NAS, NBE	0.717	1.0	0.298	0.494	IRE+1

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Conformer	Protonated atoms	Energy	Charge	$P(i)_{\text{soln}}$	$P(i)_{\text{prot}}$	MCCE
IRE+1b	NAS, N3	3.728	1.0	0.002	0.000	IRE+2
IRE+1c	NAS, N3	4.395	1.0	0.000	0.000	IRE+3
IRE+2	NAS, N3, NBE	4.412	2.0	0.000	0.000	IRE+a
Ensemble			0.302	0.300	0.494	

<i>Lapatinib (1XKK)</i>						
FMM 0	N7, N22	1.766	0.0	0.051	0.553	FMM01
FMM+1	N7(2), N22	0.030	1.0	0.949	0.447	FMM+1
FMM+2	N7(2), N22, N18	5.085	2.0	0.000	0.000	FMM+a
Ensemble			0.949	0.949	0.447	

<i>Osimertinib (4ZAU)</i>						
YY3 0	N, N3	2.299	0.0	0.021	0.008	YY301
YY3+1a	N, N2, N3	0.016	1.0	0.974	0.985	YY3+1
YY3+1b	N, N1, N3	3.310	1.0	0.004	0.003	YY3+2
YY3+1c	O, N2, N3	3.893	1.0	0.001	0.000	YY3+3
YY3+1d	O, N1, N3	4.431	1.0	0.000	0.004	YY3+4
YY3+2a	O, N, N2, N3	4.972	2.0	0.000	0.000	YY3+a
YY3+2b	N, N1, N2, N3	5.449	2.0	0.000	0.000	YY3+b
Ensemble			0.980	0.979	0.992	
JAK3: Janus kinase 3						
<i>Tofacitinib (3LXK)</i>						
MI1 0	N12	0.018	0.0	0.971	0.966	MI101
MI1+1	N12, N19	2.096	1.0	0.029	0.034	MI1+1
Ensemble			0.029	0.029	0.034	
MEK: Mitogen-activated protein kinase						
<i>Cobimetinib (4AN2)</i>						
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Conformer	Protonated atoms	Energy	Charge	$P(i)_{\text{soln}}$	$P(i)_{\text{prot}}$	MCCE
EUI 0a	O24, N9, N26	2.536	0.0	0.014	0.850	EUI01
EUI 0b	N9, N26(2)	5.637	0.0	0.000	0.000	EUI02
EUI+1	O24, N9, N26(2)	0.008	1.0	0.986	0.150	EUI+1
Ensemble			0.986	0.986	0.150	

<i>Cobimetinib (4LMN)</i>						
EUI 0a	O24, N9, N26	2.536	0.0	0.014	0.492	EUI01
EUI 0b	N9, N26(2)	5.637	0.0	0.000	0.000	EUI02
EUI+1	O24, N9, N26(2)	0.008	1.0	0.986	0.508	EUI+1
Ensemble			0.986	0.986	0.508	
MET: Tyr protein kinase						
<i>Crizotinib (2WGJ)</i>						
VGH 0	N22(2), N25	3.192	0.0	0.005	0.153	VGH01
VGH+1a	N22(2), N25(2)	0.585	1.0	0.373	0.847	VGH+1
VGH+1b	N22(2), N25, N23	3.587	1.0	0.002	0.000	VGH+2
VGH+2	N22(2), N25(2), N23	0.283	2.0	0.620	0.000	VGH+a
Ensemble			1.616	1.615	0.847	
VEGFR: Vascular endothelial growth factor receptor kinase						
<i>Axitinib (4AG8)</i>						
AXI 0a	N82, N15	0.124	0.0	0.812	0.964	AXI01
AXI 0b	N82, N14	0.993	0.0	0.187	0.036	AXI02
AXI+1	N3, N15, N82	3.790	1.0	0.002	0.000	AXI+1
Ensemble			0.002	0.002	0.000	

<i>Axitinib (4AGC)</i>						
AXI 0a	N82, N15	0.124	0.0	0.812	0.989	AXI01
AXI 0b	N82, N14	0.993	0.0	0.187	0.011	AXI02
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Conformer	Protonated atoms	Energy	Charge	$P(i)_{\text{soln}}$	$P(i)_{\text{prot}}$	MCCE
AXI+1	N3, N15, N82	3.790	1.0	0.002	0.000	AXI+1
Ensemble			0.002	0.002	0.000	

<i>Lenvatinib (3WZD)</i>						
LEV 0	N15(2), N23, N26	0.000	0.0	1.000	1.000	LEV01
LEV-1a	N15, N23, N26	5.763	-1.0	0.000	0.000	LEV-1
LEV-1b	N15, N23, N26	5.763	-1.0	0.000	0.000	LEV-2
Ensemble			0.000	0.000	0.000	

<i>Sorafenib (3WZE)</i>						
BAX 0a	N12, N14, N30	0.004	0.0	0.993	1.000	BAX01
BAX+1	N12, N14, N26, N30	3.207	1.0	0.005	0.000	BAX+1
BAX 0b	N12, N14, N26	3.620	0.0	0.003	0.000	BAX02
BAX-1	N12, N14	5.840	-1.0	0.000	0.000	BAX-1
Ensemble			0.004	0.005	0.000	

<i>Sorafenib (4ASD)</i>						
BAX 0a	N12, N14, N30	0.004	0.0	0.993	1.000	BAX01
BAX+1	N12, N14, N26, N30	3.207	1.0	0.005	0.000	BAX+1
BAX 0b	N12, N14, N26	3.620	0.0	0.003	0.000	BAX02
BAX-1	N12, N14	5.840	-1.0	0.000	0.000	BAX-1
Ensemble			0.004	0.005	0.000	

<i>Sunitinib (4AGD)</i>						
B49 0a	N23, N24, N25	2.573	0.0	0.013	0.114	B4901
B49 0b	N4, N23, N24	5.676	0.0	0.000	0.000	B4902
B49 0c	N4, N23, N25	5.692	0.0	0.000	0.000	B4903
B49+1	N4, N23, N24, N25	0.008	1.0	0.987	0.886	B49+1
Ensemble			0.987	0.987	0.886	

[†]Ponatinib binds at two distinct sites in the 3ZOS crystal structure: a buried active site and a solvent-exposed surface site, each with independent MCCE sampling.

Table S3: Outlier residues and their charge changes upon inhibitor binding

Tier	PDB	Inhibitor	ResName	Chain	ResNum	Residue	Δ Charge
strong	4MKC	Ceritinib	LYS	A	1205	LYS ⁺ A1205	−0.510
affected	4AG8	Axitinib	LYS	A	871	LYS ⁺ A0871	−0.310
affected	3UE4	Bosutinib	GLU	A	286	GLU [−] A0286	−0.270
affected	1XKK	Lapatinib	CYS	A	775	CYS [−] A0775	+0.260
affected	1XKK	Lapatinib	LYS	A	745	LYS ⁺ A0745	−0.250
affected	1XKK	Lapatinib	CYS	A	797	CYS [−] A0797	+0.230
affected	4AN2	Cobimetinib	CYS	A	207	CYS [−] A0207	−0.200
affected	4LMN	Cobimetinib	LYS	A	97	LYS ⁺ A0097	+0.200